

Math 421

Monday, October 20

Announcements

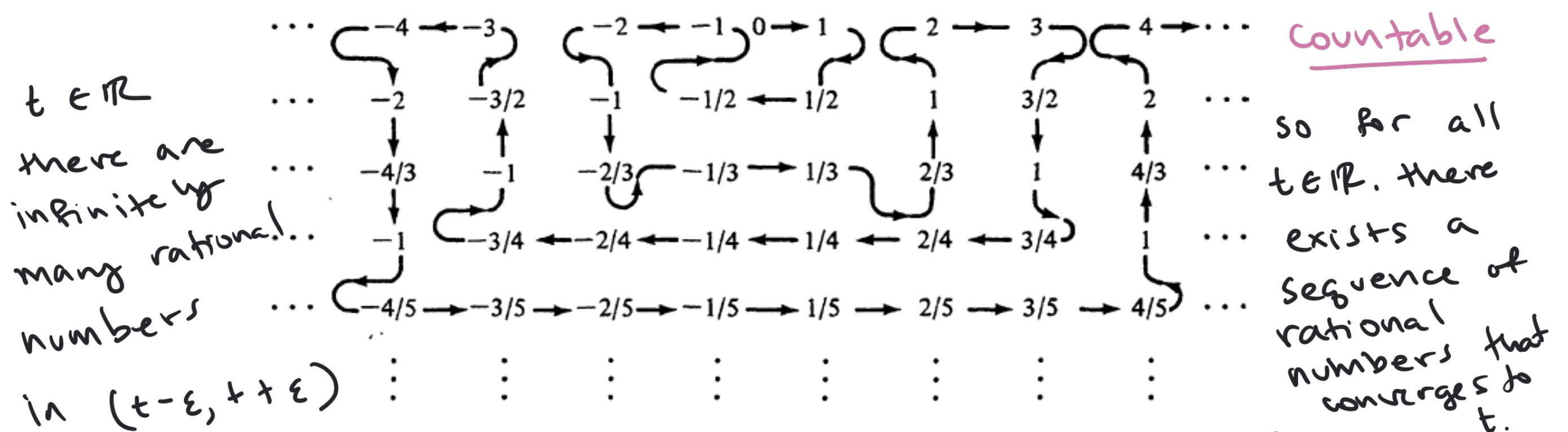
- Grace's Drop-In Hours
 - Wednesday 12-1 pm VV 311
 - Thursday 9-11 am VV B205
- MLC
 - Proof Table: M-Th 3-7:30 VV B227
 - Course Assistant: Th 4-7 Table 4
- Reading Quiz on $\liminf/\limsup$ due Wednesday
- Homework 6 Due Friday

Thm 11.2 Part 1: $t \in \mathbb{R}$. There exists a subsequence of (r_n) converging to t if and only if $\{n \in \mathbb{N} : |s_n - t| < \varepsilon\}$ for all $\varepsilon > 0$. is in

Example – Sequence of rational numbers

(r_n) – sequence consisting of all rational numbers.

This demonstrates that the rational numbers are



so for (r_n) ~~there~~ $\{n \in \mathbb{N} : |r_n - t| < \varepsilon\}$ is infinite for all $\varepsilon > 0$

if $\{n \in \mathbb{N} : s_n < \varepsilon_0\}$ is finite $\Rightarrow \{s_n < \varepsilon_0 : n \in \mathbb{N}\}$ is finite.

$$\inf \{s_n : n \in \mathbb{N}\} = \min \{s_n : n \in \mathbb{N}\}$$

$$\inf \{s_n : n \in \mathbb{N}\} \geq \inf \{s_n < \varepsilon_0 : n \in \mathbb{N}\} > 0.$$

Activity - Example.

Suppose (s_n) is a sequence of positive numbers (i.e. $s_n > 0$) such that $\inf\{s_n : n \in \mathbb{N}\} = 0$. Prove that no matter how (s_n) itself behaves, it has a subsequence converging to 0.

❖ Use Theorem 11.2 Part 1 - If t is in $\mathbb{R}$, then there is a subsequence of (s_n) converging to t if and only if the set $\{n \in \mathbb{N} : |s_n - t| < \varepsilon\}$ is infinite for all $\varepsilon > 0$.

if $\{n \in \mathbb{N} : |s_n - t| < \varepsilon\}$ is infinite for all $\varepsilon > 0$ then there exists a subsequence of (s_n) converging to t . ← want to prove

❖ Use a proof by contradiction (what are your new hypotheses?)

$\{n \in \mathbb{N} : s_n < \varepsilon_0\}$ is finite (or empty) for some $\varepsilon_0 > 0$.

- Case 1: The set is non-empty (Hint: if a set is non-empty and finite, the infimum is equal to the minimum of the set).

$$\inf \{n \in \mathbb{N} : s_n < \varepsilon_0\} = \min \{n \in \mathbb{N} : s_n < \varepsilon_0\} \quad \inf \{s_n : n \in \mathbb{N}\} = \min \{n \in \mathbb{N} : s_n < \varepsilon_0\} > 0.$$

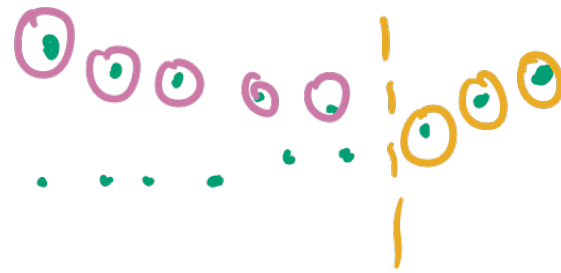
- Case 2: The set is empty. $s_n \geq \varepsilon_0$ so ε_0 is a lower bound

$$\inf \{s_n : n \in \mathbb{N}\} \geq \varepsilon_0 > 0$$

Theorems involving subsequences

Theorem 11.3: If the sequence (s_n) converges, then every subsequence converges to the same limit.

Theorem 11.4: Every sequence (s_n) has a monotonic subsequence.



Discussion Question: Convince yourselves these statements are true. It could be helpful to play around with some specific examples.